SYSTEM ARCHITECTURE

FundBridge AI · Explainable Funding Decision Support with Verified Scheme Retrieval for Micro and Small Enterprises

1 Input & Profile Capture

Business Profile
basics · story · financials
market · team · registrations

Funding Requirement
amount · purpose · type

Proposal Document
PDF / DOCX upload
validated by magic bytes, not extension

Unanswered = UNKNOWN
never defaulted in the applicant's favour

Structured fields + free text

2 Preprocessing & NLP

Document Extraction
PyMuPDF (PDF) · python-docx (DOCX)

Text Cleaning
normalise · de-hyphenate
strip control characters

Section Detection
heading heuristics
· 8 words · no mid-sentence punctuation

Section Scoring
completeness and depth per section

Python 3.12 · FastAPI · isolated service

3 Embedding & Vector Index

all-MiniLM-L6-v2
384-dimensional · L2-normalised
CPU only, no GPU required

Chunking
overlapping passages, heading retained

Vector Store
PostgreSQL double precision[]
fb_dot_product() = exact cosine

Exact search
no ANN approximation at this corpus size

pgvector unavailable for PG18 on Windows

4 Verified Scheme Corpus

Official Source Collection
government domains · robots.txt respected

Provenance Gate
rejects any record citing a non-government source

Every Field Marked
QUOTED or DERIVED
null + stated reason when unstated

PUBLISHED records only
official source URL + verification date

Synthetic demonstration records badged on screen

5 Deterministic Scoring Engines

Funding Readiness · 5 components
profile 20 · financial 25 · market 20
team 15 · proposal 20

Scheme Matching · 7 factors
industry 25 · stage 20 · type 15
funding 15 · location 10 · purpose 10
semantic relevance 5

Investor Matching · 5 factors
industry 30 · size 25 · stage 20
location 15 · funding type 10

Pure functions: no I/O, no clock, no LLM

6 Factor Resolution & Gating

Each factor resolved independently

MATCH · 1 · UNRESTRICTED · 1
the scheme states no constraint

UNKNOWN · 0
the business did not state it
forces NEEDS INFORMATION

Weight renormalisation
over the factors actually present

Blocking issue overrides the score

A high score can never come from missing data

7 Grounded Retrieval (RAG)

Query embedding
the same 384-d model

Cosine similarity
raw threshold · 0.35
PUBLISHED documents only

Below threshold · returns nothing
and states why it cannot answer

LLM phrasing
Groq · openai/gpt-oss-120b
explains a decision, never makes one

Retrieved text is data, never instructions

8 Explainable Output

Score with its composition
every factor, its weight and its reason

Four honest statuses
potentially eligible · needs information
review required · likely not eligible

Named gaps
the exact fields that would raise each score

Citations
official URL + verification date

modelVersion + configVersion
stamped on every derived row

Decision support · never a guarantee of funding

Processing block · Model / weighted stage · Permissive outcome · Conservative guard · Data flow

Explainable by Construction · **Conservative Under Uncertainty** · **Provenance-Tracked Scheme Data** · **Grounded Generation (RAG)**